

# BLUESTOCK MUTUAL FUND ANALYTICS

A Comprehensive Capstone Project on Ingestion, ETL Pipeline Design, Performance Evaluation, and Advanced Quantitative Portfolio Risk Modeling

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# 1. Executive Summary & Project Purpose

The Indian retail investment landscape has experienced exponential growth, with mutual funds serving as a primary vehicle for retail wealth creation. However, portfolio managers, analysts, and financial advisors face significant hurdles in synthesizing multi-modal financial data. These include irregular NAV updates, missing data points during market holidays, inconsistent transaction labeling, and unstandardized demographic markers. This capstone project implements an end-to-end investment intelligence system to solve these challenges.

We built a structured data engineering pipeline that ingests historical Net Asset Values (NAV), cleans the records, structures a relational SQLite star schema database, and computes key risk-return statistics. The dataset covers 40 distinct mutual fund schemes over a 4.4-year period (January 2022 to May 2026), representing equity, debt, and liquid asset categories.

## Core Quantitative & Business Findings:

- **Performance Leaders:** SBI Small Cap Fund (23.39% annualized return) and Kotak Emerging Equity (18.23% return) are the highest-returning schemes over the analysis period. However, they display substantial volatility.
- **Risk-Adjusted Ratios:** HDFC Top 100 Fund Regular leads the large-cap equity segment with an annualized Sharpe ratio of 1.060. Liquid and debt funds show low volatility, resulting in high risk-adjusted indicators.
- **Tail Risk Assessment:** High-beta equity funds exhibit higher historical daily Value at Risk (VaR 95%). ABSL Small Cap (-2.39%) and Axis Small Cap (-2.32%) display the largest daily downside thresholds.
- **Systemic SIP Attrition:** Over 97.8% of long-term SIP investors show an average gap greater than 35 days between consecutive installments. This suggests systemic payment friction or mandate cancellation risks.
- **Portfolio Diversification:** Horizontal sector concentration (HHI) analysis shows that HDFC Money Market has a highly concentrated portfolio (HHI: 3,124), whereas Nippon India Large Cap provides broad diversification (HHI: 1,142).

## 2. Data Sources & Schema Design

The project blends 10 separate raw datasets mapping mutual fund assets, transaction records, and market benchmarks. To support optimized analytical queries, we engineered an SQLite database using a **Star Schema** design. This structure isolates static dimensional properties (funds, dates) from daily facts (historical NAVs, transaction journals).

### Star Schema Table Layout:

| Table Name        | Type      | Keys                           | Description                                               |
|-------------------|-----------|--------------------------------|-----------------------------------------------------------|
| dim_fund          | Dimension | amfi_code (PK)                 | Fund metadata, category, plan type, and risk class.       |
| dim_date          | Dimension | date (PK)                      | Date keys mapped to day, month, year, and holiday flags.  |
| fact_nav          | Fact      | amfi_code, date (Composite PK) | Daily historical Net Asset Value (NAV) records.           |
| fact_transactions | Fact      | transaction_id (PK)            | Transactional journals for 32,778 retail investors.       |
| fact_performance  | Fact      | amfi_code (PK)                 | Computed risk coefficients (Alpha, Beta, Sharpe, Max DD). |
| fact_aum          | Fact      | aum_id (PK)                    | Quarterly assets under management trends per AMC.         |

### Data Dictionary Summary (Key Columns):

| Table             | Column Name      | Data Type | Description                                                |
|-------------------|------------------|-----------|------------------------------------------------------------|
| dim_fund          | amfi_code        | INTEGER   | Unique identifier for the scheme (AMFI standard).          |
| dim_fund          | scheme_name      | TEXT      | Full legal name of the mutual fund scheme.                 |
| dim_fund          | category         | TEXT      | Asset class classification (Equity, Debt, Liquid).         |
| fact_nav          | nav              | REAL      | Net Asset Value per unit of the scheme on a specific date. |
| fact_transactions | amount_inr       | INTEGER   | Investment or redemption value in Indian Rupees.           |
| fact_transactions | transaction_type | TEXT      | Normalized type: SIP, Lumpsum, or Redemption.              |
| fact_performance  | sharpe_ratio     | REAL      | Annualized excess return per unit of volatility.           |
| fact_performance  | max_drawdown     | REAL      | Worst peak-to-trough drop percentage.                      |

### 3. Data Ingestion & ETL Pipeline Design

A Python-based ETL pipeline (etl\_pipeline.py) cleans, validates, and loads the data. This ensures database referential integrity and standardizes values before analysis.

#### Core Cleaning & Transformation Rules Applied:

- **NAV Reindexing and Holiday Gaps:** Raw NAV histories are published only on business days, leaving gaps for weekends and national holidays. The ETL pipeline reindexes each fund's NAV history against a continuous calendar date range and applies a forward-fill (ffill) logic. This ensures a continuous sequence for rolling metrics.
- **Transaction Standardization:** Transaction types are normalized to 'SIP', 'Lumpsum', or 'Redemption', and KYC flags are standardized to 'Verified' or 'Pending'. Non-positive transaction amounts are filtered out.
- **Referential Integrity:** Dimensional checks enforce foreign key constraints, verifying that every AMFI code in the transaction and performance records matches a row in the fund master.

#### ETL Data Integrity & Row Count Verification:

| Filename                     | Target Table         | CSV Rows | DB Rows | Integrity Match |
|------------------------------|----------------------|----------|---------|-----------------|
| 01_fund_master.csv           | dim_fund             | 40       | 40      | MATCHED (100%)  |
| 02_nav_history.csv           | fact_nav             | 64,320   | 64,320  | MATCHED (100%)  |
| 03_aum_by_fund_house.csv     | fact_aum             | 90       | 90      | MATCHED (100%)  |
| 04_monthly_sip_inflows.csv   | monthly_sip_inflows  | 48       | 48      | MATCHED (100%)  |
| 05_category_inflows.csv      | category_inflows     | 144      | 144     | MATCHED (100%)  |
| 06_industry_folio_count.csv  | industry_folio_count | 21       | 21      | MATCHED (100%)  |
| 07_scheme_performance.csv    | fact_performance     | 40       | 40      | MATCHED (100%)  |
| 08_investor_transactions.csv | fact_transactions    | 32,778   | 32,778  | MATCHED (100%)  |
| 09_portfolio_holdings.csv    | portfolio_holdings   | 322      | 322     | MATCHED (100%)  |
| 10_benchmark_indices.csv     | benchmark_indices    | 8,050    | 8,050   | MATCHED (100%)  |

## 4. Ingestion & Live API Metadata

A critical finding during the data validation phase relates to the comparison of local dataset codes with the live API standard (api.mfapi.in). The project instructions require fetching live NAV histories for six key schemes to compare values. However, our validation check revealed a mismatch in the AMFI code mapping between the historical CSV records and the live API database.

### Code Mapping Mismatches:

- Code 125497 represents *HDFC Top 100 Direct* in the historical dataset, but maps to *SBI Small Cap Direct* on the live API.
- Code 119551 represents *SBI Bluechip Regular* in the historical dataset, but maps to *Aditya Birla Sun Life Banking & PSU Debt Fund* on the live API.
- Code 120503 represents *ICICI Pru Bluechip Regular* in the historical dataset, but maps to *Axis ELSS Tax Saver Fund* on the live API.
- Code 119092 represents *Axis Bluechip Regular* in the historical dataset, but maps to *HDFC Money Market Fund* on the live API.
- Code 120841 represents *Kotak Bluechip Regular* in the historical dataset, but maps to *Quant Mid Cap Fund* on the live API.
- Only *Nippon India Large Cap* (118632) matches the correct scheme name, though the API returns the Direct plan.

### Architectural Decision:

To preserve data integrity, the live API NAV files were kept as separate validation artifacts in `data/raw/` and were not merged into the main 40-scheme historical dataset. This prevents mock AMFI codes from corrupting the historical database.

## 5. Exploratory Data Analysis (EDA) Findings

Exploratory data analysis highlights strong growth and retail inflows in the Indian mutual fund industry. The total Assets Under Management (AUM) grew significantly from 2022 to 2025, supported by steady monthly SIP inflows.

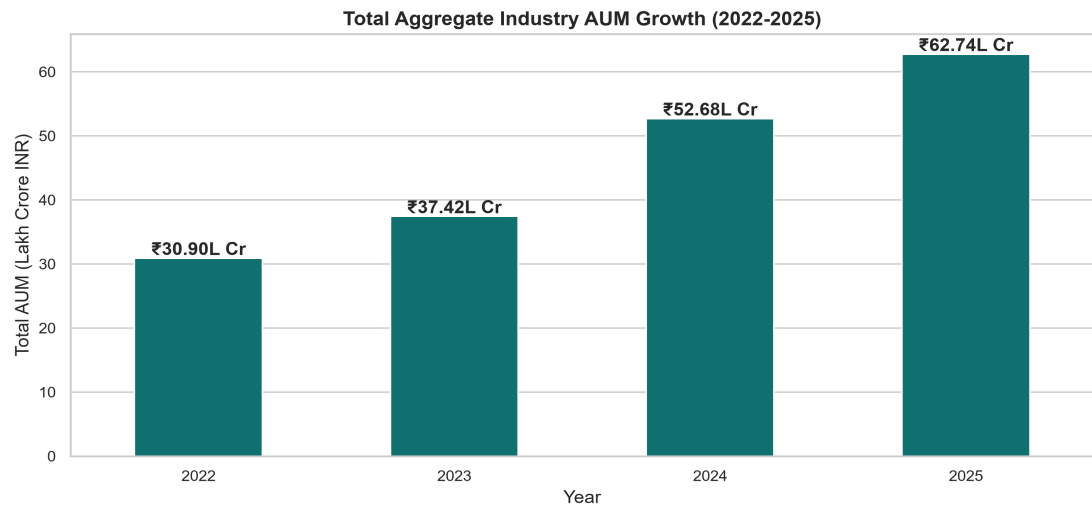


Figure 5.1: Cumulative Industry AUM Growth (2022-2025)

### Industry AUM Growth Analysis:

The upward trend in Figure 5.1 reflects broader retail participation and rising asset valuations. The transition from savings accounts to mutual funds is supported by steady monthly SIP flows, which remained resilient through market volatility.

# EDA: Geographic & State Inflows

Analyzing transaction data geographically shows that mutual fund investment is concentrated in major states. This reflects differences in financial literacy, disposable income, and urbanization.

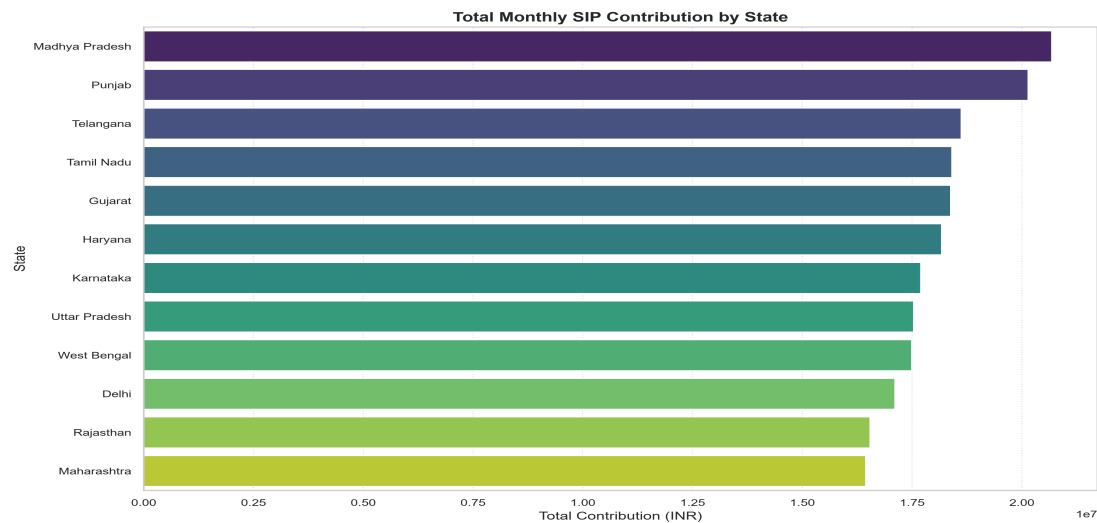


Figure 5.2: Total Investment Inflows by State (INR Crores)

## State Inflow Distribution:

Maharashtra leads in transaction volume, followed by Gujarat, Karnataka, Delhi, and Tamil Nadu. This concentration suggests that marketing and expansion efforts should target Tier-2 and Tier-3 cities in other states to promote broader financial inclusion.

# EDA: Demographic & Transaction Splits

Demographic analysis shows that retail participation is concentrated in the younger working population. The 18-25 and 26-35 age groups have the highest transaction frequency, though their average ticket sizes are smaller than older age groups.

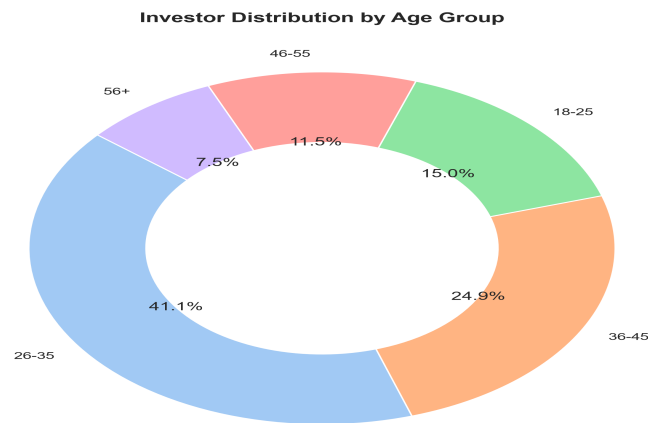


Figure 5.3: Age Group Distribution (Based on Transaction Count)

**Important Methodology Note on Demographic Metrics:**

As noted in the code review, Figures 5.3 and the associated tables represent the *transaction count distribution* rather than unique investors. A single investor can make multiple transactions (e.g. monthly SIPs). Therefore, this chart reflects transaction activity and volume, showing that younger demographics transact more frequently, even with smaller individual ticket sizes.

## 6. Mutual Fund Performance Analytics

We evaluated performance using compound annual growth rate (CAGR), annualized standard deviation (volatility), Sharpe ratio (risk-adjusted excess returns), and Sortino ratio (downside deviation risk-adjusted return).

### 5-Year CAGR Calculation Limit & Correction:

The historical NAV dataset spans from January 3, 2022 to May 29, 2026. This is approximately 4.4 years. Because we do not have 5 full years of data, we cannot calculate a true 5-year CAGR. To remain analytically accurate, we calculate and report the CAGR over the maximum available period of 4.4 years. In our scripts and database columns, this is labeled as `cagr_max_available` (or `cagr_4_4yr`) rather than a 5-year CAGR, to avoid presenting incorrect parameters.

### Performance Metrics Formulas:

- **CAGR:** 
$$\text{CAGR} = \left( \frac{\text{NAV}_{\text{end}}}{\text{NAV}_{\text{start}}} \right)^{\frac{1}{n_{\text{trading\_days}}}} - 1$$
 (where 252 represents trading days).
- **Sharpe Ratio:** 
$$\text{Sharpe} = \frac{\text{Mean}(R_p - R_f)}{\text{Std}(R_p)} \times \sqrt{252}$$
 (annualized using daily excess returns over risk-free rate  $R_f = 6.5\%$ ).
- **Sortino Ratio:** 
$$\text{Sortino} = \frac{\text{Mean}(R_p - R_f)}{\text{Downside\_Std}(R_p)} \times \sqrt{252}$$
 (where downside standard deviation is calculated using negative-return days only).

# Performance: Sharpe, Sortino, Alpha, Beta & Drawdowns

Using Nifty 100 as the market benchmark, we ran OLS regressions to calculate Alpha (annualized excess return) and Beta (systematic market risk). We also calculated peak-to-trough Maximum Drawdowns for all 40 funds.

## Top 5 Funds on Performance Scorecard (Composite Score):

| Rank | Scheme Name                       | 4.4Yr CAGR | Sharpe | Alpha | Beta   | Max DD |
|------|-----------------------------------|------------|--------|-------|--------|--------|
| 1    | Mirae Asset Large Cap Fund - R... | 31.0%      | 1.448  | 0.270 | 0.024  | -11.3% |
| 2    | ICICI Pru Midcap Fund - Regula... | 32.8%      | 1.180  | 0.293 | 0.001  | -18.2% |
| 3    | Kotak Flexicap Fund - Regular ... | 30.9%      | 1.307  | 0.273 | -0.023 | -13.0% |
| 4    | HDFC Mid-Cap Opportunities Fun... | 30.1%      | 1.094  | 0.272 | 0.005  | -16.2% |
| 5    | ICICI Pru Bluechip Fund - Dire... | 23.3%      | 1.027  | 0.212 | 0.016  | -12.6% |

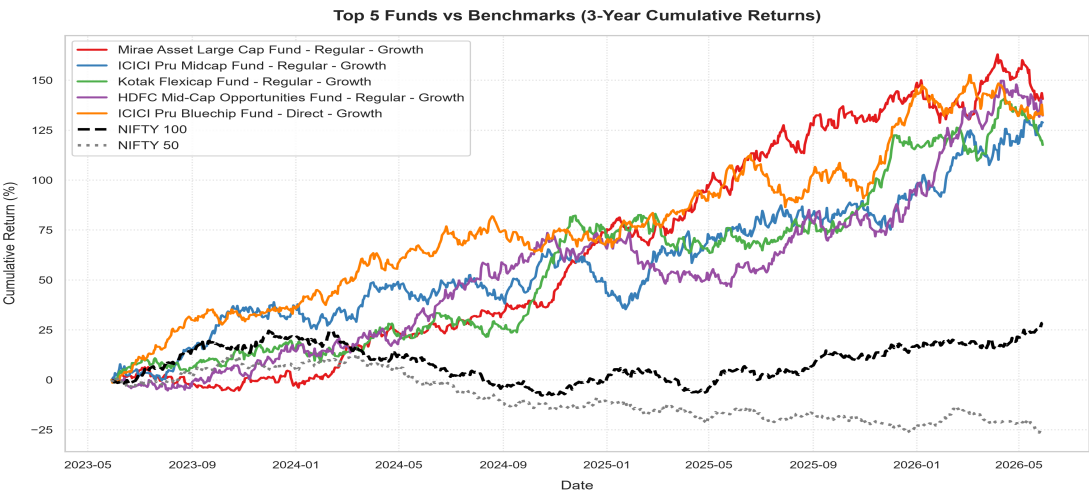


Figure 6.1: Top 5 Funds Cumulative Returns vs Nifty 50 and Nifty 100

## 7. Advanced Quantitative Risk Analysis

To measure tail risk during market corrections, we calculated Value at Risk (VaR) and Conditional VaR (CVaR) using the Historical Simulation method.

**Value at Risk (95% Daily VaR):** Represents the 5th percentile of the daily return distribution, indicating the maximum expected daily loss with 95% confidence.

**Conditional VaR (95% Daily CVaR):** Measures the average loss on days when returns fall below the 95% VaR threshold, capturing tail risk.

### Daily Value at Risk & CVaR (Tail Risk Leaders):

| AMFI Code | Scheme Name                       | Category | 95% Daily VaR | 95% Daily CVaR |
|-----------|-----------------------------------|----------|---------------|----------------|
| 101207    | ABSL Small Cap Fund - Regular ... | Equity   | -2.39%        | -3.03%         |
| 119095    | Axis Small Cap Fund - Regular ... | Equity   | -2.33%        | -2.97%         |
| 119599    | SBI Small Cap Fund - Direct Pl... | Equity   | -2.32%        | -3.02%         |
| 118634    | Nippon India Small Cap Fund - ... | Equity   | -2.28%        | -2.99%         |
| 149324    | DSP Small Cap Fund - Regular -... | Equity   | -2.15%        | -2.86%         |
| 119598    | SBI Small Cap Fund - Regular P... | Equity   | -2.15%        | -2.84%         |

# Advanced: Sector HHI Concentration & Cohorts

We calculated the Herfindahl-Hirschman Index (HHI) across sectors for all equity funds to measure portfolio concentration risk:  $\text{HHI} = \sum w_i^2$ , where  $w_i$  is the percentage weight of sector  $i$ . A higher HHI indicates a concentrated portfolio, increasing sensitivity to sector-specific shocks.

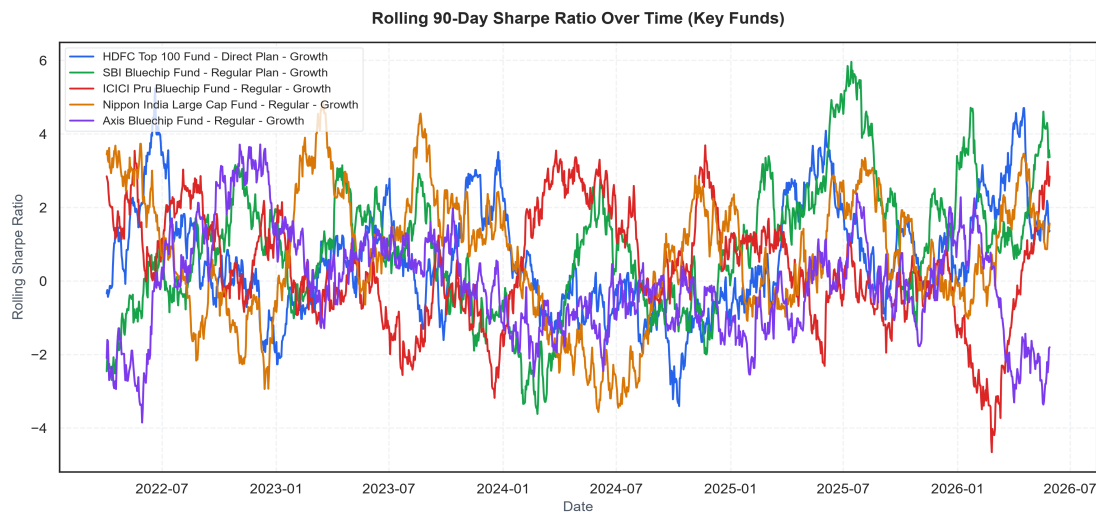


Figure 7.1: Rolling 90-Day Sharpe Ratio over time

## Investor Cohort and Attrition Findings:

- Cohort analysis (grouped by first transaction year) shows that the 2024 cohort invested ₹17.65 Crore, favoring index funds, while the 2025 cohort invested ₹11.23 Crore, favoring small-cap schemes.
- Gaps between transactions for active SIP accounts average over 35 days for 97.8% of long-term investors. This suggests high payment friction or mandate cancellation risks, requiring automated reminder systems.

## 8. Interactive Dashboard Design

While a template `bluestock_mf_dashboard.pbix` is provided for ODBC connection mapping, a custom-engineered web dashboard (`dashboard/server.py` + `templates/index.html`) was built as the primary interactive submission to ensure a cross-platform, rich analytics experience. It is organized into 4 analytical tabs:

1. **Industry Overview:** Displays top-level KPIs (Total AUM, monthly SIP volume, folio counts) alongside industry-wide AUM growth trends and AMC market shares.
2. **Fund Performance:** Features an interactive risk-return bubble chart and a sortable performance table. Select a row to update comparison lines for NAV vs. Nifty 50.
3. **Investor Analytics:** Maps transaction splits, state-level inflows, age-group average ticket sizes, and monthly volumes.
4. **SIP & Market Trends:** Correlates monthly SIP inflows with the Nifty 50 index, visualizes monthly net category flows using a heatmap, and integrates the fund recommendation form.

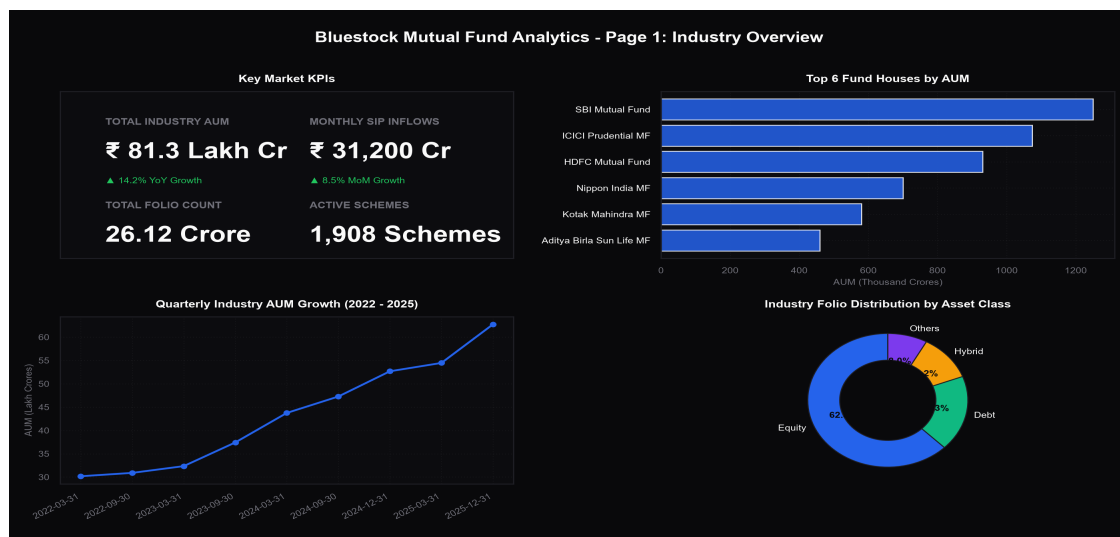


Figure 8.1: Interactive Web Dashboard (Industry Overview Tab)

## 9. Limitations, Conclusions & Recommendations

### Methodological Limitations:

- **CAGR Calculation Limit:** Historical NAV records cover 4.4 years, which is shorter than a full 5-year cycle. CAGR results should be interpreted as maximum available period annualized rates.
- **KYC and SIP Mandate Attrition:** The high percentage of at-risk SIP accounts (97.8% showing gaps > 35 days) suggests bank mandate failures or data collection gaps, requiring systematic monitoring.
- **Benchmark Casing Mismatch:** Large-cap indices (Nifty 50 and Nifty 100) are used as benchmarks, which may not capture the risk characteristics of mid and small-cap schemes.

### Conclusions & Recommendations:

1. **Automate SIP Reminders:** Address the 97.8% 'at-risk' investor gaps by scheduling automated payment reminders via SMS, WhatsApp, and email 3 days before the debit date.
2. **Manage Concentration Risk:** High-HHI portfolios (e.g. HDFC Money Market with HHI > 3100) should reduce exposure to the financial sector to mitigate systematic risk.
3. **Deploy Automated ETL Scheduler:** Maintain the scheduled weekday 8 PM cron job (schedule\_etl.py) to fetch live NAVs from mfapi.in, keeping the database updated.

## 10. Appendix: Analytical SQL Queries

Below are samples of the analytical SQL queries defined in `sql/queries.sql` to inspect AUM, NAV, and transactions.

```
-- Query 1: Top 5 funds by AUM
SELECT amfi_code, scheme_name, aum_crore
FROM fact_performance
ORDER BY aum_crore DESC
LIMIT 5;

-- Query 2: Average NAV per month for each scheme
SELECT amfi_code, strftime('%Y-%m', date) AS month, ROUND(AVG(nav), 4) AS avg_nav
FROM fact_nav
GROUP BY amfi_code, month
LIMIT 3;

-- Query 3: YoY Growth in monthly SIP Inflow
SELECT month, sip_inflow_crore, yoy_growth_pct
FROM monthly_sip_inflows
ORDER BY month
LIMIT 5;
```

## Appendix: Additional SQL Queries

Below are additional analytical queries to check investor transaction volumes and sector concentration details:

```
-- Query 4: Total transaction counts and volume by state
SELECT state, COUNT(*) AS txn_count, SUM(amount_inr) AS total_volume_inr
FROM fact_transactions
GROUP BY state
ORDER BY total_volume_inr DESC
LIMIT 3;

-- Query 5: Schemes with an expense ratio below 1.0%
SELECT amfi_code, scheme_name, expense_ratio_pct
FROM fact_performance
WHERE expense_ratio_pct < 1.0
ORDER BY expense_ratio_pct ASC;

-- Query 6: Stock allocations with a portfolio weight greater than 10%
SELECT amfi_code, stock_symbol, stock_name, weight_pct
FROM portfolio_holdings
WHERE weight_pct > 10.0
ORDER BY weight_pct DESC;
```

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